

Critical Reflection of the Global Digital Payment Gender Gap: What Drives It?

1 Introduction

Digitalisation is transforming personal and business payments, across both private and public sectors. Digital payment platforms have become an essential part of financial infrastructure across the world. Indeed, financial technology has the potential to create a more inclusive, resilient, and sustainable world (World Bank, 2023). However, inequality of access to financial technology persists, and limits its potential to promote financial inclusion. In particular, a gender gap in access to financial services is systematic and persistent. This is due to deficiencies in regulation, as well as inequalities in income and access to mobile technology, with 9% more males owning accounts than females worldwide (Delechat et al., 2018). This gender gap in access to financial services restricts economic opportunities for women, and reinforces pre-existing socio-economic inequalities in society.

To explore these gender disparities, we used the Global Findex Dataset published in 2021, which provides data on payment, savings, borrowing, income, and access to financial technology for those over 14 at the country level, including data collected from previous years. This report focuses on digital payments and mobile money in 2017 and 2021. Using this subset, we examine the gender gaps in digital finance across economies, paying particular attention to disparities between developed and developing countries, and discover the related drivers of financial inclusion.

Our central research question is:

What are the key country-level drivers of the digital payment gender gap?

We define the *digital payment gender gap* as the difference, in percentage points, between the share of men and the share of women who made or received a digital payment in a given year. Potential drivers considered include basic account ownership, mobile-money account ownership, and other financial behaviours such as receiving wages or government payments, saving, borrowing, and sending remittances.

1. Our analysis proceeds in three steps. First, we describe how digital payments and gender gaps evolved between 2017 and 2021 globally and across regions.
2. Second, we explore correlations between the digital payment gender gap and other indicators.
3. Third, we fit a linear regression model (for 2021 data) to quantify which country-level factors are most strongly associated with the digital payment gender gap.

The report concludes with a discussion of limitations of our model, confounding factors, and possible policy implications.

2 Preparation

2.1 Data selection and structure

The original **DatabankWide.xlsx** file contains 658 observations and 1,232 variables. Observations correspond to either individual countries or aggregate regions (e.g. “World”, “High income”) and span survey years 2011, 2014, 2017 and 2021. Variables include indicators on account ownership, saving, borrowing, digital payments, mobile money, and demographics. All outcome variables are stored as proportions between 0 and 1.

For this project we restricted attention to country-level observations in 2017 and 2021, the two most recent survey waves with broad coverage of digital-finance questions. Aggregate rows such as “World”, “Euro area”, or “High income” were dropped to avoid double counting. After these filters, the working dataset contained 290 country-year observations.

2.2 Variable construction

The main response variable is the **digital payment gender gap**:

$$\begin{aligned} \text{DigitalPaymentGenderGap}_c \\ = \text{Digital payment use, men}_c - \text{Digital payment use, women}_c, \end{aligned}$$

where digital payment use is the share of adults (15+) who made or received a digital payment in the previous year in country c .

Analogous gender-gap variables were created for several related indicators, including:

- Overall account gender gap: difference between men and women in “having an account at a financial institution or through a mobile-money provider”.
- Mobile-money account gender gap.
- Gender gaps in having received wages, government transfers, domestic remittances, saved any money, borrowed any money, and being able to raise emergency funds in seven days.

Each gender gap is measured in proportion units (between -1 and 1). For interpretation in the text, we usually express these as percentage points.

We also defined two non-gap predictors:

- Mobile-money penetration: share of adults with a mobile-money account.
- Mobile-phone ownership: share of adults who own a mobile phone.

Finally, an income-group level variable encodes the World Bank income classification as an ordered scale from 1 (= low income) to 4 (= high income). Region and income-group labels from the Findex metadata were retained to allow regional comparisons.

2.3 Handling missing data

Not all variables are available for all countries. For descriptive statistics, we used all available observations in each year, so the number of countries, and the particular countries observed, can vary across indicators. For the regression analysis, we required complete data on the response and all explanatory variables in 2021. This list-wise deletion yields a modelling sample of 65 countries. The choice avoids making additional assumptions about the missing values but may bias our model towards countries with better data availability; we return to this limitation in Section 5.

Throughout the analysis the data are treated as country-level point estimates. We do not use Findex survey weights or standard errors, so sampling uncertainty within each country is ignored. The results should therefore be interpreted as describing patterns across country averages rather than individual-level relationships.

3 Exploration

3.1 Global trends 2017–2021

Global averages show strong growth in digital payment use for both men and women between 2017 and 2021. The mean share of women who made or received a digital payment rose from roughly 50% in 2017 to 61% in 2021. For men, the corresponding increase was from 57% to 68%. Global account ownership also increased from around 62% to 71%, and mobile-money account coverage from about 14% to 23%.

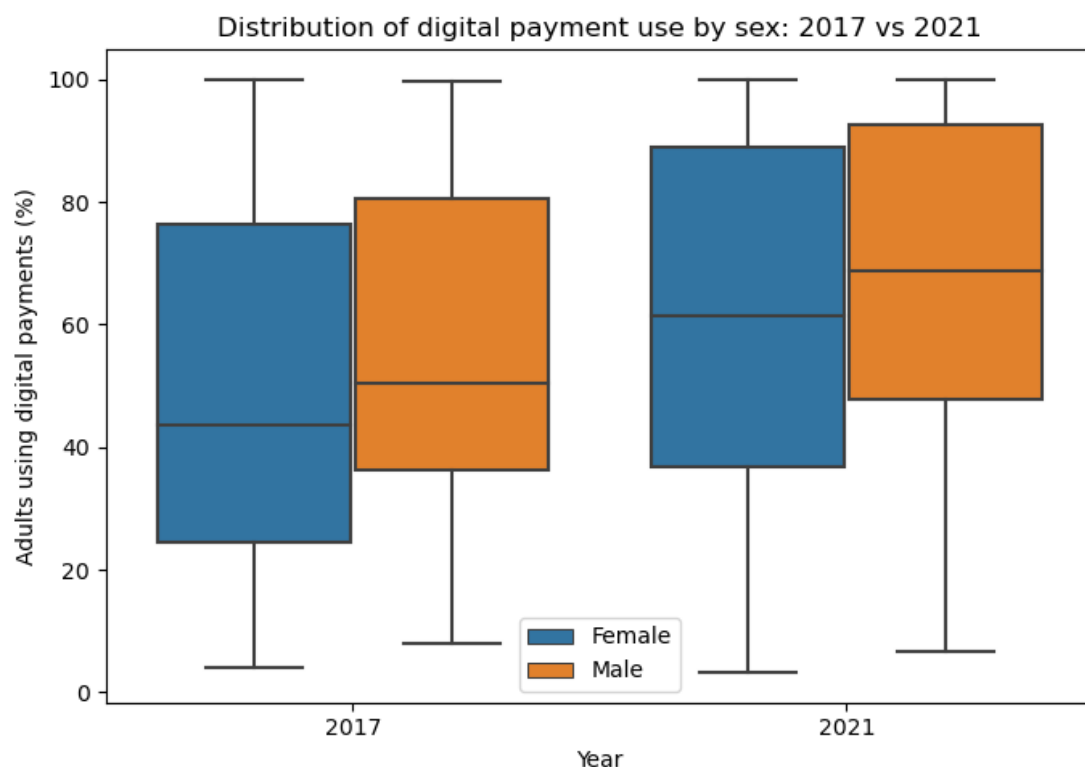


Figure 1: Share of adults (15+) who made or received a digital payment in the past year, by sex, global average, 2017 vs. 2021.

As shown in the figure above, both men and women increased their use of digital payments. Despite these improvements, gender differences remain. The global digital payment gender gap declined only modestly, from about 7.1 percentage points in 2017 to 6.6 points in 2021. The account ownership gender gap narrowed more noticeably, from 7.1 to 6.0 percentage points. Thus, while women are increasingly using digital payments, they still lag behind men by a non-trivial margin.

From the figure below, we compare the gender gaps in account ownership and digital payments.

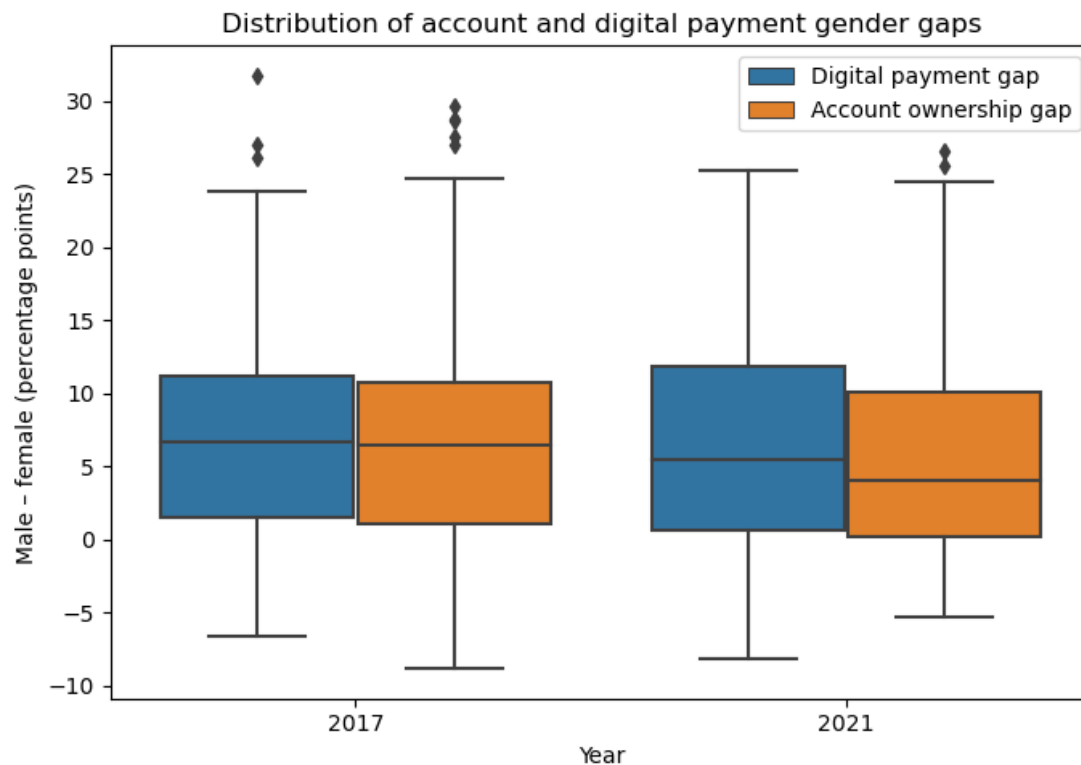


Figure 2: Global average gender gaps (male – female, in percentage points) in account ownership and digital payment use, 2017 vs 2021.

3.2 Regional patterns

There is substantial heterogeneity across regions. In 2021, adults in high-income economies have the highest digital payment usage: on average, about 93% of adults made or received a digital payment, and the gender gap is small at roughly 1.5 percentage points. Account ownership is almost universal in this group.

The figure above highlights the contrast between high-income economies and regions such as South Asia and MENA.

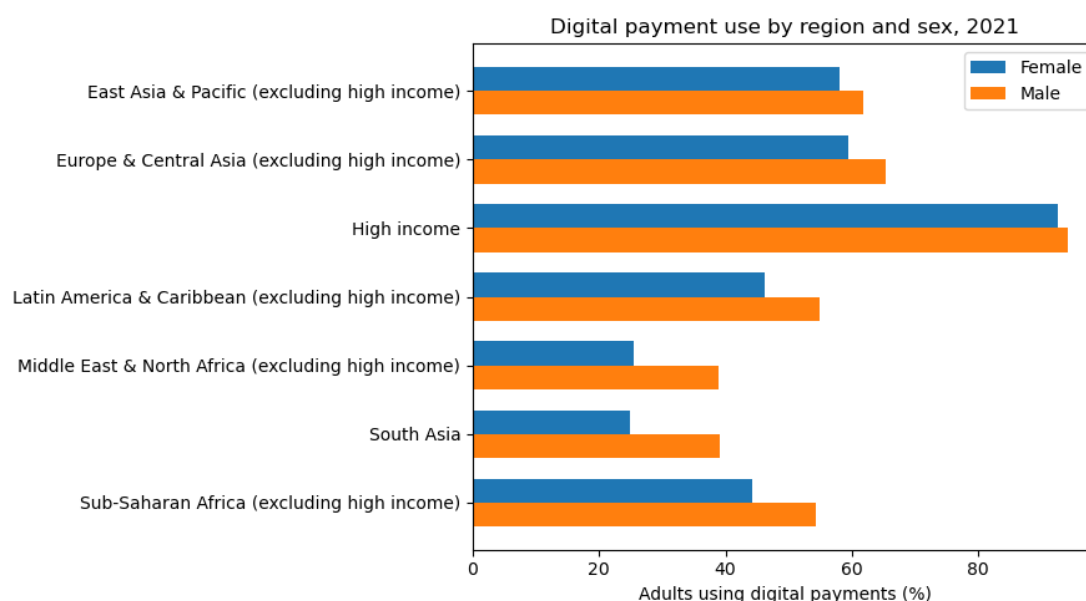


Figure 3: Share of adults (15+) who made or received a digital payment in the past year, by region and sex, 2021.

At the other extreme, South Asia and the Middle East & North Africa (MENA) exhibit both low levels of digital use and large gender gaps. In South Asia, only about 32% of adults used digital payments in 2021, and the average digital payment gender gap is around 14.3 percentage points. MENA has a similar average gap of 13.5 points and low mobile-money penetration. Sub-Saharan Africa stands out for its relatively high mobile-money penetration: about 36% of adults have a mobile-money account, but the digital payment gender gap still averages around 10 percentage points.

Comparing 2017 and 2021, most regions experienced higher digital usage but only modest reductions in gender gaps. High-income economies reduced their digital gap from about 3.6 to 1.5 percentage points. In contrast, the gap in South Asia barely changed (from 14.7 to 14.3 points), and in Sub-Saharan Africa it increased slightly (from 9.4 to 10.2 points). This suggests that expanding digital finance does not automatically translate into more gender-equal use. Women are still less likely to own mobile phones and even if they do, such phones are usually not smart phones (GSMA, 2025). Plus, despite rising financial advancements, the gender gap remains because women are still less likely to learn about digital skills and basic financial concepts (Marconi et al., 2025).

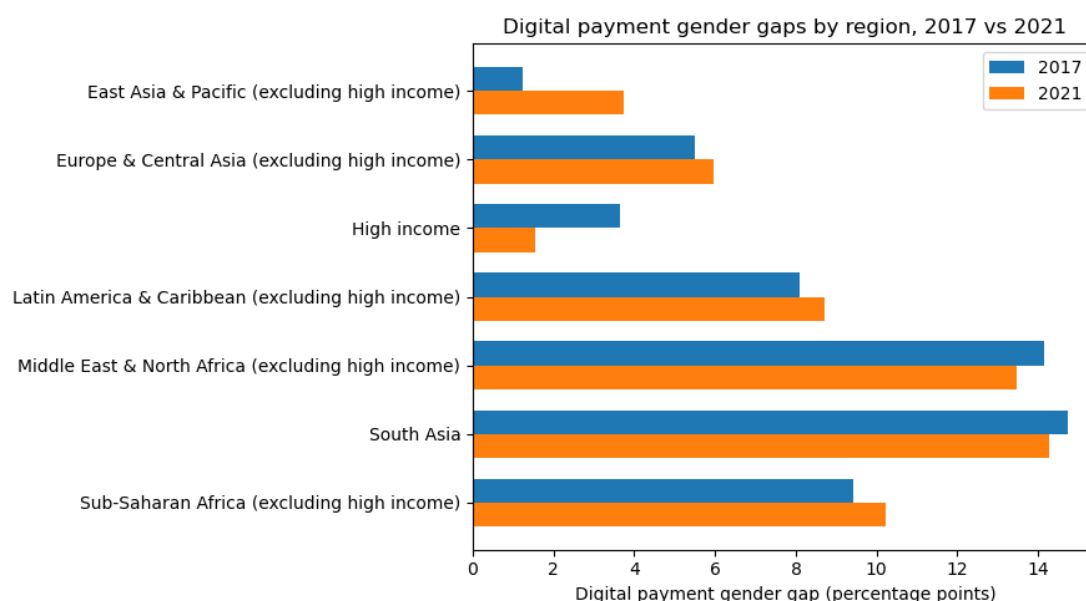


Figure 4: Average digital payment gender gap (male – female percentage points) by region, 2017 and 2021).

3.3 Summary statistics and correlations

Summary statistics for 2021 further highlight the heterogeneity across countries. The mean digital payment gender gap is about 6.6 percentage points, with a standard deviation of 7.0 points and a range from –8.2 (where women use digital payments more than men) to +25.3 points. The mean account ownership gender gap is around 6.0 points, and the mean mobile-money gender gap, among countries with such accounts, is about 5.5 points. Mobile-phone ownership is high on average (about 88% of adults), but lowest value is only 32%.

Variable	mean	std	min	max
Made or received a digital payment, female (% age 15+)	0.612	0.287	0.032	1.000
Made or received a digital payment, male (% age 15+)	0.678	0.250	0.066	1.000
DigitalPaymentGenderGap	0.066	0.070	-0.082	0.253
Account, female (% age 15+)	0.676	0.266	0.042	1.000
Account, male (% age 15+)	0.736	0.227	0.077	1.000
OverallAccountGenderGap	0.060	0.071	-0.053	0.266
Mobile money account (% age 15+)	0.227	0.169	0.000	0.687
MobileMoneyGenderGap	0.055	0.050	-0.049	0.211
Own a mobile phone (% age 15+)	0.877	0.118	0.322	1.000

Table 1: Descriptive Statistics of key country-level indicators, 2021.

Simple correlations in the 2021 cross-section reveal strong relationships. The digital payment gender gap has a correlation of 0.92 with the account gender gap and 0.66 with the mobile-money account gender gap. Gaps in receiving remittances, government payments, and wages digitally are also positively correlated with the digital gap, though less strongly (between 0.53 and 0.56). In contrast, the correlation between the digital gap and overall mobile-money penetration is slightly negative (around -0.07). These patterns suggest that inequalities in account ownership, especially mobile-money accounts, may be central drivers of gender gaps in digital payments.

4 Analysis

4.1 Model specification

To quantify these associations, we estimate an ordinary least squares (OLS) regression using the 65 countries with complete data in 2021. The dependent variable is the digital payment gender gap (i.e. the percentage point difference in digital payment usage between genders). The main explanatory variables are:

- Overall account gender gap;
- Mobile-money account gender gap;
- Mobile-money penetration (share of adults with a mobile-money account);
- Mobile-phone ownership (share of adults owning a mobile phone); and
- Income-group level (1 = low income, ..., 4 = high income).

This specification aims to separate the role of basic account access, mobile-money-specific access, and broader structural factors such as income level and phone ownership. All continuous variables are left in their original proportional units. The model is:

$$DigGap_c = \beta_0 + \beta_1 AccGap_c + \beta_2 MMGap_c + \beta_3 MMlevel_c + \beta_4 Phone_c + \beta_5 Income_c + \varepsilon_c.$$

4.2 Regression results

The OLS results show a very good fit: the model explains about **82%** of the cross-country variation in the digital payment gender gap ($R^2 = 0.817$). The estimated coefficients are:

- Constant: -0.024 (p = 0.37)
- Account gender gap: 0.657 (standard error 0.072, p < 0.001)
- Mobile-money gender gap: 0.382 (0.108, p = 0.001)
- Mobile-money account: -0.056 (0.024, p = 0.020)
- Mobile phone ownership: 0.075 (0.041, p = 0.072)
- Income-group level: -0.004 (0.007, p = 0.62).

	Coefficient	Std. Error	t-stat	p-value
const	-0.0243	0.0269	-0.9046	0.3694
OverallAccountGenderGap	0.6568	0.0724	9.0771	0.0000
MobileMoneyGenderGap	0.3822	0.1083	3.5281	0.0008
Mobile money account (% age 15+)	-0.0564	0.0236	-2.3907	0.0200
Own a mobile phone (% age 15+)	0.0750	0.0409	1.8338	0.0717
IncomeGroupLevel	-0.0036	0.0073	-0.4939	0.6232

Table 2: OLS regression of the digital payment gender gap, 2021.

The account gender gap has the largest and most precisely estimated effect. The coefficient of 0.657 implies that, holding other variables constant, a 10-percentage-point larger gap in account ownership (Suri & Jack, 2016) is associated with a 6.6-percentage-point larger digital payment gender gap. This confirms the descriptive impression that countries where women are less likely to hold any account are also those where they are furthest behind in using digital payment

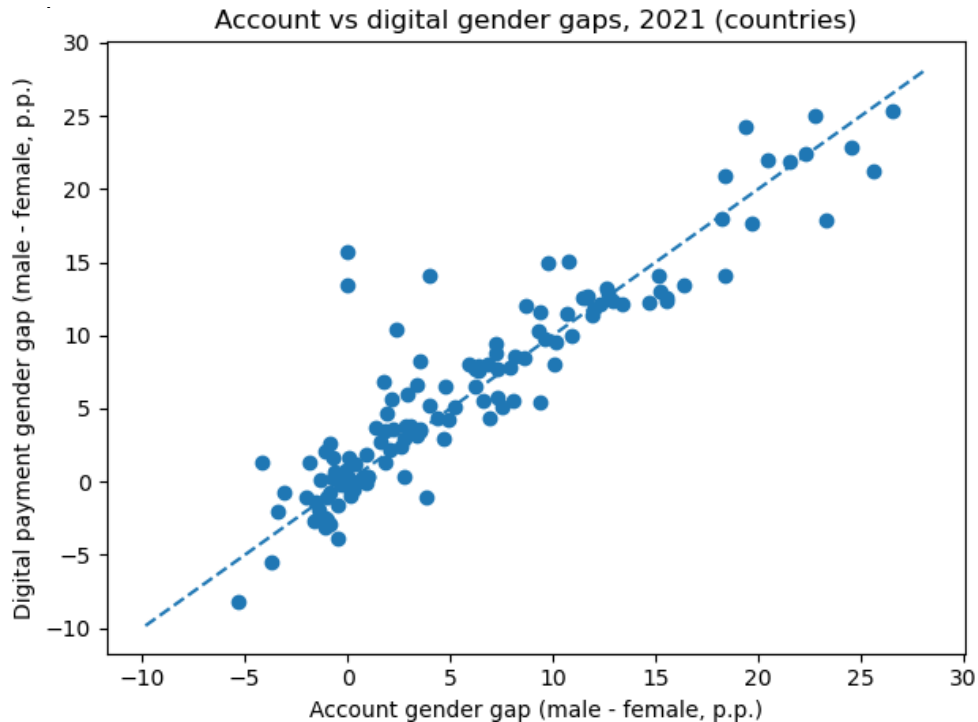


Figure 5: Scatterplot of account gender gap vs. digital payment gender gap (male–female percentage points) for 2021, with 45-degree line.

Graphically, Figure 5 shows that the digital payment gender gap is strongly positively associated with the account gender gap.

The mobile-money account gender gap also has a sizeable and statistically significant association: a 10 percentage-point larger gap in mobile-money account ownership corresponds to about a 3.8 percentage-point larger digital payment gender gap. This effect remains even after controlling for the basic account gap, suggesting that mobile money introduces an additional dimension of gender inequality.

Interestingly, the coefficient on overall mobile-money penetration is negative and statistically significant. For a given set of gender gaps, countries where mobile-money accounts are widely used have somewhat smaller digital payment gender gaps. One interpretation is that broad, mature mobile-money ecosystems may make it easier for women to adopt digital payments — perhaps through lower transaction costs, more agent networks, or targeted products — even if they still lag men in account ownership (Suri & Jack, 2016).

In contrast, once account-related gaps are controlled for, mobile-phone ownership (which itself correlates with mobile-money account ownership) and income group add little explanatory power: their coefficients are relatively small and statistically insignificant at conventional levels. This suggests that simply being richer or having high phone coverage is not sufficient to guarantee small digital gender gaps; what matters more is whether women are actually able to obtain and use financial accounts, especially mobile-money accounts.

4.3 Model assessment

Residual diagnostics indicate that the linear specification is broadly adequate. Residuals are centred around zero with no strong non-linear patterns in residual-versus-fitted plots, suggesting approximate linearity and homoscedasticity. A normal Q–Q plot shows some deviations in the tails, consistent with the presence of a few countries with unusually high or low gaps, but no extreme outliers.

Normal Q–Q plot of residuals: OLS model for digital payment gender gap

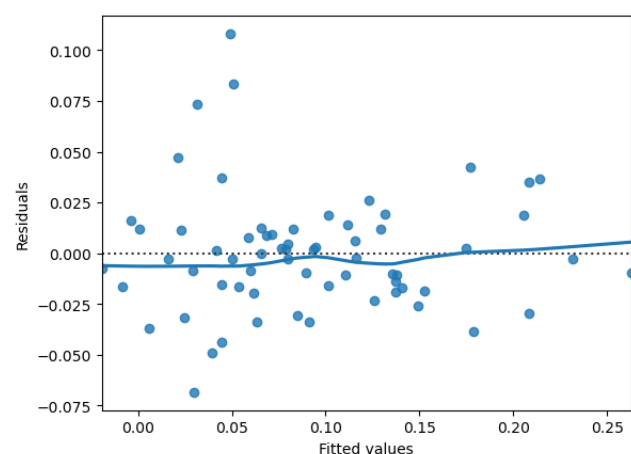
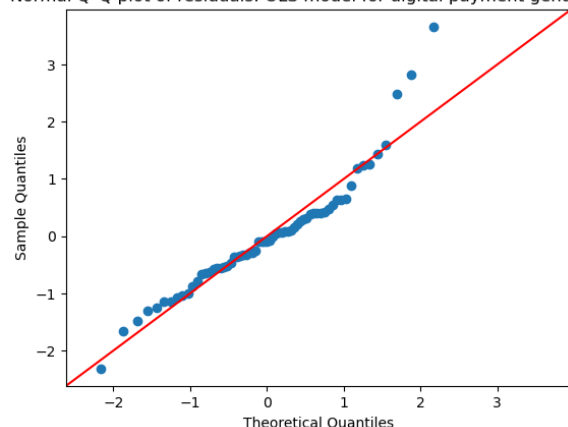


Figure 6: Residuals versus fitted values from the OLS regression in Table 2.

Some caveats remain. The account and mobile-money gender gaps are themselves highly correlated, so multicollinearity could inflate standard errors. Most importantly, the analysis is cross-sectional and observational: the coefficients should be interpreted as statistical associations, and not as causality.

5 Presentation and Conclusion

The descriptive analysis shows that digital payments have expanded rapidly for both men and women. However, gender gaps narrowed only modestly over the same period. Regional disparities are stark: high-income economies combine nearly universal digital payment use with very small gaps, whereas South Asia and MENA exhibit both low female usage and large persistent gaps. Sub-Saharan Africa has relatively high mobile-money penetration but still sizeable gender gaps.

The regression results provide a clear answer to the main question. The digital payment gender gap is primarily driven by two factors:

1. The gender gap in basic account ownership; and
2. The gender gap in mobile-money account ownership.

Together with overall mobile-money penetration, these variables explain more than four-fifths of the cross-country variation in the digital payment gender gap. Other variables such as income group and mobile-phone ownership are much less informative once account-related gaps are taken into account. In short, where women are as likely as men to hold accounts — especially mobile-money accounts — the digital payment gender gap is small (World Bank, 2022), even when overall usage is moderate. Where women remain excluded from accounts and mobile money, expanding digital finance can leave them further behind.

The investigation has several limitations:

- First, it relies on country-level averages, which may mask substantial within-country heterogeneity by age, income, or location.
- Second, the analysis does not incorporate survey weights or measurement error, and treats the published Findex estimates as exact.

- Third, the cross-sectional design precludes strong causal claims: unobserved factors such as legal restrictions on women's financial rights, social norms, or female labour-force participation may drive both account gaps and digital payment gaps (World Bank, 2024).
- Finally, list-wise deletion of missing data reduces the sample for modelling and may bias results towards better-measured countries.

Despite these caveats, the findings suggest clear policy directions. Efforts to close the digital payment gender gap should prioritise removing barriers to women's account ownership, including simplifying documentation requirements, improving access to IDs, and addressing legal or social constraints that limit women's control over financial assets.

At the same time, regulators and providers should ensure that the rapid expansion of mobile-money services is inclusive of women, for example through agent networks in areas where women live and work, gender-sensitive product design, and targeted financial-literacy programmes.

Future work could extend this analysis by incorporating external indicators (such as education, internet access, or gender-equality indices), using microdata to model individual-level adoption, or exploiting the full panel of Findex waves to study how changes in policies and infrastructure affect gender gaps over time.

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